



**The Faculty of Information and Communication Technology
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**Project 1: Database Planning and Requirement Analysis
E-Commerce: IKEA**

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ITCS442 Relational Database Design
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Business Domain and Problem statement

Real Business Company: IKEA (Customer and Order Management System)

IKEA Thailand provides an omnichannel retail service including online ordering, click & collect, delivery service, installation service, assembly service, etc. Customers can use the IKEA website to track their orders and schedule to access the services, while IKEA staff are responsible for the process such as preparing orders, scheduling deliveries, and arranging booking services. However, operational data for each section is separated across different systems and workflows, such as tracking delivery, furniture assembly services, and other services. Each service has different conditions depending on the specifications of the product, resulting in an increased amount of data and the need for continuous status updates.

Problem statement

This project proposes the development of a centralized database application that combines order fulfillment, service booking, and return management into a single system. This system aims to reduce operational errors of scattered data and improve data connectivity across different services.

Declaration of AI tool usage : AI Level 2

This project used AI tools (ChatGPT, Gemini) as a supporting assistant for ideas, improving English writing (paraphrasing, grammar correction, and clarity), and organizing report structure. All database design decisions (e.g. key characteristics, mission statement, mission object, scope and boundary diagram, business function, and requirement specification) were reviewed and verified. The final analysis, implementation, and report were completed and finalized by team members. We take full responsibility for the accuracy, originality, and integrity of the submitted work.

As-Is Analysis

Key characteristics of the current situation (Target Users)

Roles	Tasks	Details
Customer	end-buyers	Browses catalog, pays for orders, and tracks assembly bookings.
Operational Staff	Stock Management	Updates bin locations and adjusts stock levels per store.
Service Team	Operation Management	Manages the calendar for delivery trucks and assembly crews.
System Admin	IT support	Manages user permissions and monitors database backups.
Customer Support Staff	Case Handling	Handles inquiries about orders, refunds/returns, and shipping or scheduling.

Core entities

Data	Description	Attributes
Product	Stores information about IKEA products, to make it easier for customers to browse or purchase for a product.	name, product cereal number, dimensions, weight, flat-pack count.
Customer	Stores customers personal detail, so customers can place orders, receive the services, and track their order.	name, address, phone number, email, IKEA Family membership status.
Order	Stores each online purchase record, including order date, total price, and order status.	date, status, total price.
Payment	Stores the payment record that gets information from the order, but only focusing on the payment that received from the bank and payment history.	payment id, order id, product name, customer name, payment history, payment status.
Membership	Stores IKEA Family membership information that is linked to a customer, for giving special discounts to a customer.	membership id, customer name, membership tier.
Staff	Stores staff information for the system to assign the correct responsible for stock handling, order picking, delivery, and assembly services.	name, address, phone number, email.

Key Business Rules :

1. An assembly service can not be booked unless the order contains at least one furniture category item.
2. The system must check stock availability in the specific store selected by the customer before confirming a "Click & Collect" order.
3. A "Refund Request" can only be initiated if the Order status is delivered and within 365 days.
4. An order cannot be completed if the product is "Out of Stock" at the selected store.

Mission Statement and Object

Mission Statement

The mission of this database application is to build a centralized database application for supporting the front-end and managing the service in IKEA Thailand into a single connected platform. By gathering service data that is distributed across separate services, this system has a goal to reduce operational errors of the service data in workflows, improve the consistency and accuracy of real-time data across different services or departments, and support smoother coordination between customers and staff.

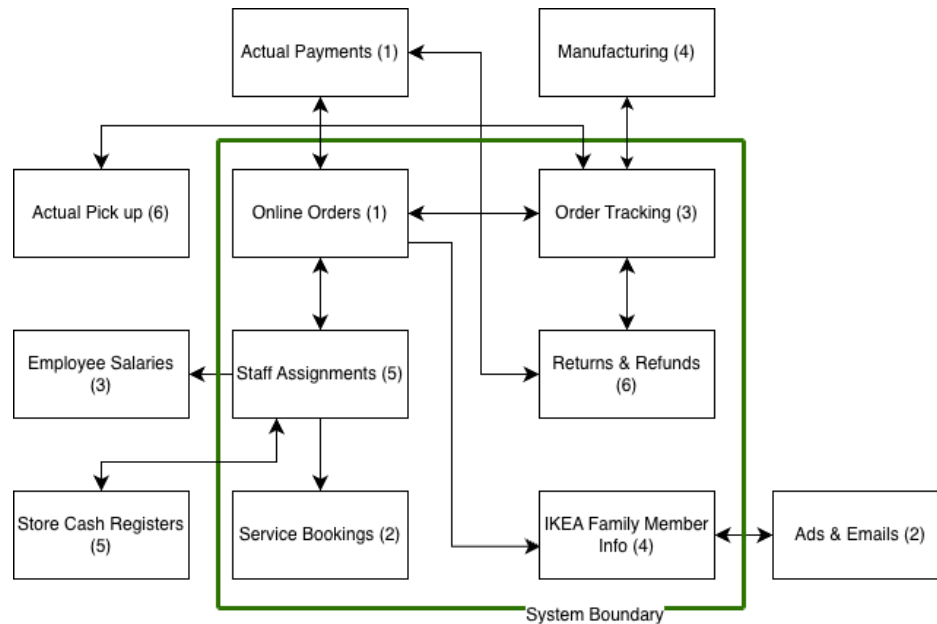
This development of this database application will enhance service reliability, improve customer experience, and develop the connectivity of database management systems.

Mission Object

1	Combine all service data into a single system	to reduce inconsistency across multiple IKEA service processes.
2	Keep product and stock availability data	to ensure ordering and inventory information has the same up-to-date information.
3	Manage the complete order process in one place	to be able to see the initial process to the completed process.
4	Combine delivery and pickup scheduling data	to ensure that staff and customers have the same confirmed plan.
5	Keep order tracking status logs	to ensure the timeline of updated information history is completed.
6	Centralize service booking management	to combine schedules of booked service in the same order record.
7	Merge returns and refunds system management	to ensure the status of the transaction has completely followed.

Scope and boundary

This section defines what our database will do and what it will stay away from



As shown in Figure 1, the system boundary of IKEA Online Store Database system is covered by 6 scope modules. These scope include **Online Order (1)**, which manages the core function for recording customer purchases and order details, and integrates with **Service Booking (2)** for schedule delivery and assembly services. To make the order progress and inventory management accurate, the system also includes **Online Tracking (3)**, which monitors the order status and product availability. Moreover, the system also stores **IKEA Family Member Info (4)** to support the part of member discounts and purchase history, and manages **Staff Assignment (5)** to make an accurate assignment for staff to delivery drivers or assembly workers to a specific orders or service bookings, and records **Returns & Refunds (6)** to track the return requests and refund status.

Overall, these six modules work together as an integrated database system that supports online order processing, service management, inventory and order tracking, staff assignment, and return & refunds handling. Ensuing a smooth and consistent flow of the data across IKEA's Online store operations.

In contrast, the boxes outside the green boundary are out of scope systems that are not implemented inside this database application. These external systems including **Actual Payments (1)** are processed by external payment providers, and the IKEA system only receives the payment result messages. **Ads & Emails (2)** is an external marketing system that may use IKEA Family Member data for communication. **Employee Salaries (3)** is about Human Resource functions that are not included in the system. **Manufacturing (4)** is not covered in this database. **Store Cash Registers (5)** is used for in-store transactions. **Actual Pick up (6)** will operate when customers collect items at the store.

Overall, the figure clarifies that this project focuses on centralizing IKEA Thailand's core service data for online operations while treating in-store activities as an external system.

1. Stock and Product management: To make sure the online store and the physical warehouse have the exact same information so customers don't buy items that are out of stock.

- 2. Ordering and buying management:** To move the customer's choice from the website into a real "work order" and ensure IKEA receives the correct payment.

- 3. Service and Staff Scheduling:** To connect the physical product with the human help (delivery and assembly) needed to get the furniture ready in the customer's home.

- 4. Customer and Member Care (IKEA Family):** To build a relationship with shoppers by remembering their details and rewarding them with member-only benefits.

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- ```
graph LR
 subgraph IKEA_E-commerce [IKEA E-commerce]
 direction TB
 subgraph Actors
 Customer
 Operational_Staff[Operational Staff]
 Service_Team[Service Team]
 System_Admin[System Admin]
 Customer_Support[Customer Support Staff]
 end
 subgraph Use_Cases
 direction TB
 UC1((Browse Catalog))
 UC2((Place Online Order))
 UC3((Manage Profile))
 UC4((Request Return/Refund))
 UC5((Track Order/Booking))
 UC6((Update Stock Levels))
 UC7((Confirm Order Picking))
 UC8((Review Return Requests))
 UC9((Manage Service Calendar))
 UC10((Update Job Status))
 UC11((View Delivery Details))
 UC12((Manage User Permissions))
 UC13((Monitor Database Backups))
 UC14((Maintain System Logs))
 UC15((Receive Inquiry))
 UC16((View Order History))
 UC17((Resolve Issue))
 end
 subgraph External_Systems
 direction TB
 Banking_System((Banking System))
 Cloud_Provider[(Cloud Database Provider)]
 Legal_Authority((Legal/Tax Authority))
 end
 UC1 --> UC2
 UC3 --> UC2
 UC4 --> UC2
 UC5 --> UC2
 UC6 --> UC7
 UC8 --> UC7
 UC9 --> UC7
 UC10 --> UC7
 UC11 --> UC12
 UC13 --> UC14
 UC15 --> UC14
 UC16 --> UC17
 UC17 --> UC18
 end
 Customer --> UC1
 Customer --> UC3
 Customer --> UC4
 Customer --> UC5
 Operational_Staff --> UC6
 Operational_Staff --> UC8
 Operational_Staff --> UC10
 Service_Team --> UC9
 Service_Team --> UC11
 System_Admin --> UC12
 System_Admin --> UC13
 System_Admin --> UC14
 Customer_Support --> UC15
 Customer_Support --> UC16
 Customer_Support --> UC17

 UC2 -.->|<<extend>>| UC2_1((Apply Membership Discount))
 UC2_1 -.->|<<include>>| UC2_2((Process Payment))
 UC2_2 --> Banking_System

 UC7 -.->|<<include>>| UC7_1((Sync Inventory))
 UC7_1 --> Cloud_Provider

 UC14 --> Legal_Authority
```

## Mapping of Major User Views and Operations

| Data \ User | Access Type | Customer | Operation Staff | Service Team | System Admin | Customer Support Staff |
|-------------|-------------|----------|-----------------|--------------|--------------|------------------------|
| Product     | Maintain    |          | X               |              | X            |                        |
|             | Query       | X        | X               | X            | X            |                        |
|             | Report      | X        | X               | X            | X            | X                      |
| Customer    | Maintain    |          |                 | X            | X            | X                      |
|             | Query       |          | X               |              | X            | X                      |
|             | Report      |          | X               | X            | X            | X                      |
| Order       | Maintain    |          | X               |              | X            |                        |
|             | Query       | X        | X               | X            | X            |                        |
|             | Report      | X        | X               | X            | X            | X                      |
| Payment     | Maintain    |          | X               |              | X            |                        |
|             | Query       |          |                 |              | X            |                        |
|             | Report      | X        | X               | X            | X            | X                      |
| Membership  | Maintain    | X        | X               |              | X            |                        |
|             | Query       | X        | X               |              | X            |                        |
|             | Report      |          |                 |              | X            | X                      |
| Staff       | Maintain    |          |                 |              | X            |                        |
|             | Query       |          |                 | X            | X            |                        |
|             | Report      | X        | X               |              | X            |                        |



# Users' Requirements Specification

## Data Requirements

### Product

The data stored on each product includes product name, product cereal number, dimensions, and weight so the customers can browse and purchase items correctly. The system must store inventory data such as stock quantity and stock status for checking stock availability for a "Click & Collect" order and must prevent operational error in the ordering process when the product is out of stock.

### Order

The data stored on each online order includes date, status, and total price. The database also needs to keep order tracking status logs to support the timeline and history of updates and manage the order process from the initial process to completed process.

### Customer

The data stored on each customer includes name, address, phone number, email, and status of IKEA Family membership so the customers can place orders, receive services, and track their orders. Customer data must support privacy and security controls and apply with PDPA to handle the personal information.

### Payment

The data stored on each payment include payment id, order id, product name, customer name, payment history, and payment status to make the accurate tracking and verification of payment transactions and order processing. Each payment must be linked to one specific order. Payment status must be recorded as "successful", "failed", or "refunded". Payment information must not be changed after the payment status has been finalized. Each payment must record only one payment method.

### Membership

The data stored on each membership includes membership id, customer name, and membership tier. Each membership must have a unique membership ID for identification. Each membership must have only one tier at any given time. A customer may hold only one active membership at a time. Membership information must be retained in the system after the membership has expired.

### Staff

The data stored on each staff includes name, address, phone number, and email. Each staff member must be assigned to only one role at any given time. Staff contact information must be up to date for operational purposes. Each staff email address must be unique in the system. Each staff information may only be updated or changed by the staff member who owns the record.

## Transaction Requirements

### Data Approach

- Enter the details of a new online order that is created from a customer cart.
- Enter the details of a payment result that is returned from the payment of the specific order.
- Enter the details of a service booking that is linked to a correct order.
- Enter the details of a staff assignment to an order or service job.

### Data Update / Deletion

- Update / delete the details of an online order.
- Update the payment status of an order based on the payment status and store payment history.
- Update the stock quantity of a product after an order is being placed, confirmed, and update the stock when the item is being returned and refunded.
- Update / delete staff assignment details for an order, service job and booking.
- Update the return and refund status and delete the duplicated or invalid return request.

### Data Queries

- List payment records and order details for a given order ID.
- List the current stock availability for a product and list the low-stock and out-of-stock products.
- List the service booking by date / time and view booking details.
- List staff assignments by staff ID or time period.
- List returns and refunds for a given customer or order and list the status summary.

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## System Requirements

### 1. Storage and Growth

- **Initial Size:** 150 GB to store all IKEA product photos, stock lists, and member info.
- **Yearly Growth:** The data will grow by 30 GB every year as we add new furniture and more orders.

### 2. Performance (Speed)

- **Peak Times:** During big IKEA sales, the system must handle 1,000 orders every minute without crashing.
- **Search Speed:** When a customer looks for a product, the results must show up in less than 1.5 seconds.

### 3. Access and Networking

- **Shared Access:** Up to 300 staff and 5,000 customers can use the system at the same time.
- **Connection:** Stores in each store connect to a Cloud Database to see the same stock counts instantly.

### 4. Security and Backups

- **Privacy:** Customer names and addresses are encrypted so hackers cannot read them.
- **Staff Roles:** A delivery driver can see a customer's address but cannot see how much they paid.
- **Backups:** We save a new copy of the data every hour so we never lose an order if the power goes out.

### 5. Legal and History

- **History:** We keep all receipts for 10 years to follow Thai tax and return laws.
- **Privacy Law:** The system follows PDPA rules to keep customer phone numbers and emails private.

## Reference

Formal UML Logic (Standard Rules) : <https://www.omg.org/spec/UML/>

*Industry Reference: IKEA Thailand – Customer Service and Returns Policy:*  
<https://www.ikea.com/th/en/customer-service/>.

*ITCS442 Relational Database Design lect 1-3 Faculty of Information and Communication Technology Mahidol university : Instructor: Asst.Prof. Dr. Charnyote Pluempitiwiriyawej.*

*AI level 2 from Google Gemini :* <https://gemini.google.com>

*Operations management of IKEA SlideShare.com :*  
<https://www.slideshare.net/slideshow/operations-management-of-ikea/187051571>

*User registration form* <https://family.ikea.co.th/registration>

*Employee application:*  
<https://jobs.smartrecruiters.com/IkanoRetail/744000102921265-business-analyst-finance-business-partner-commercial-finance-ikea-sukhumvit-and-ikea-bangna?trid=c39a8cee-af0e-4db1-9034-acbba85a4a4d>

*Delivery service report and Tracking online orders report:*  
<https://www.ikea.com/mx/en/customer-service/track-manage-order/>

*IKEA: A Case Study Through the Lens of Algorithms and Data Structures:*  
<https://medium.com/@harshada.deshingkar22/ikea-a-case-study-through-the-lens-of-algorithms-and-data-structures-af8de39e47b7>

*Case Study Ikea:* <https://www.scribd.com/document/468047669/Case-Study-Ikea>

*Use case Diagram from lucid.app:*  
[https://lucid.app/lucidchart/010d2b9a-366d-4ae5-b0b8-7c791433a3d5/edit?invitationId=inv\\_3603f791-aa59-4658-8a96-66a3d22bf86a](https://lucid.app/lucidchart/010d2b9a-366d-4ae5-b0b8-7c791433a3d5/edit?invitationId=inv_3603f791-aa59-4658-8a96-66a3d22bf86a)

*System and boundary diagram from draw.io:*  
[https://drive.google.com/file/d/1UqdvIA\\_ltw7rvLBpeILYLLjwyldW-z\\_3/view?usp=sharing](https://drive.google.com/file/d/1UqdvIA_ltw7rvLBpeILYLLjwyldW-z_3/view?usp=sharing)